Zilin Dai

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Education

Harvard University, Cambridge, MA — GPA: 3.96

– Computational Science and Engineering M.E., Full Fellowship by Amazon *August 2025 – May 2027**

Worcester Polytechnic Institute (WPI), Worcester, MA — GPA: 3.97

– Computer Science B.S. *August 2021 – May 2025*

Related Courses: Machine Learning, Principles of Autonomy and Decision Making, Generative AI, On-Device Deep Learning, Deep Learning, Digital Image Processing, Algorithms, Operating Systems, Modeling and Data Analysis, Human-Computer Interaction, AI for Robotics, Principles of Physics-Mechanics, Object-Oriented Programming

Selected Publications

Shuo Xing*, **Zilin Dai***, Chengyuan Qian, Fangzhou Lin, Wenjing Chen, Ping He, Pan Lu, Alvaro Velasquez, Mohit Bansal, Zhengzhong Tu. *The Missing Primitive: Diagnosing and Repairing Mathematical Reasoning in Large Language Models.* arXiv 2026.

Fangzhou Lin*, **Zilin Dai***, Shuo Xing, Peiran Li, Siyuan Yang, Qianwen Ge, Kazunori Yamada, Yang Zhou, Ziming Zhang, Haichong Zhang, Zhengzhong Tu. *AdaptFuse: Training-Free Sequential Preference Inference via Externalized Bayesian Computation.* arXiv 2026.

Zilin Dai, Lehong Wang, Fangzhou Lin, Ziming Zhang, Wang Lu. *Frozen Language Anchors as Noise-Robust Priors for Domain Generalization under Label Noise: An Empirical Study.* In Proceedings of ECCV E.T. 2026.

Yi Yang*, Xiang Fei*, Lehong Wang*, Chenhao Li, **Zilin Dai**, Henry Kou, Howie Choset, Lu Li. *DeformX: A Versatile Co-Simulation Framework for Deformable Linear Objects.* Accepted to IROS 2026.

Zilin Dai, Lehong Wang, Yi Yang, Xiang Fei. *WireSeg-32K: A Physics-Grounded Synthetic Dataset for Wire Instance Segmentation.* Accepted to Synthetic Data for Computer Vision @ CVPR 2026.

Zilin Dai*, Fangzhou Lin*, Rigved Sanku*, Songlin Hou, Kazunori D Yamada, Haichong K. Zhang, Ziming Zhang. *A Strong View-Free Baseline Approach for Single-View Image Guided Point Cloud Completion.* arXiv.

Xinlu He, Yiwen Guan, Badrivishal Paurana, **Zilin Dai**, Jacob Whitehill. *Interactive Real-Time Speaker Diarization Correction with Human Feedback.* Submitted to ACCL 2026.

Lehong Wang, **Zilin Dai** et al. *From Flat to Form-Fitting: A Computational Geometry Approach to 3D Conformal Electronics Design and Rapid Prototyping.* IEEE CASE 2025.

Zilin Dai*, Yijia Wu*, Haotian Liu, Lehong Wang, Markus P. Nemitz. 2024. *Vision-based FDM printing for fabricating airtight soft actuators.* IEEE International Conference on Soft Robotics 2024 (Oral Presentation).

Zilin Dai*, Lehong Wang*, Markus P. Nemitz. 2024. *Fully FDM-Printed Soft Hand Exoskeleton.* IEEE International Conference on Robotics and Automation (ICRA): Emerging Technologies in Smart Exoskeleton Systems 2024.

Zilin Dai, Andrew McReynolds, Jacob Richard Whitehill. 2023. *In Search of Negative Moments: Multi-Modal Analysis of Teacher Negativity in Classroom Observation Videos.* International Conference on Educational Data Mining 2023.

Intellectual Property

Dongsheng Dai, Lynn Shan, **Zilin Dai**. 2024. *Circular RNAS and Preparation Methods Thereof.* International Publication Number: WO 2024/051842 A1

Dongsheng Dai, Lynn Shan, **Zilin Dai**. 2024. *Circular RNAS for Expressing Urate Oxidase, and Preparation Methods and Uses Thereof.* International Publication Number: WO 2024/051842 A1

Experience

Robotic Data Science Intern, Amazon, Bellevue, WA *May 2026 – Present*

– Designing and prototyping an E2E AI-generated warehouse blueprint system that combines classical mathematical optimization with VLM-based verification, augmented by pretrained RL policies for efficient iterative refinement.

– Scoped an open-ended problem statement into a phased roadmap with milestones and evaluation criteria, aligning scientists and engineering stakeholders on priorities and deliverables.

Master's Thesis under Prof. Debora Marks, Harvard Medical School, MA *September 2026 – Present*

– Formulating protein-binding RNA design as conditional sequence generation, coupling a pretrained protein encoder to an in-house RNA foundation model through a learned conditioning interface; comparing autoregressive, masked language modeling, and masked discrete diffusion objectives.

– Designing a three-tier evaluation from local optimization to unconstrained generation, with wet-lab dissociation constants as terminal validation, yielding a benchmark and affinity labels that the field currently lacks.

Research Student under Prof. Yilun Du, Harvard University, MA *January 2026 – September 2026*

– Leading a research project on compositional world models for long-horizon robot manipulation, studying how structured latent representations can improve prediction, planning, and policy learning in multi-stage tasks.

AI for Drug Discovery Intern, Pfizer, Cambridge, MA *June 2025 – Dec 2025*

- Led pretraining of multi-modal Mamba and Transformer-based protein language model with domain-specific tokenizer, performed distributed training across internal HPC and AWS for scalable and fast training iteration.
- Built a high-throughput retrieval pipeline for clustered antibody datasets, shrinking per-sample access from intractable runtimes to milliseconds, enabling real-time training and analysis. Workflow and code deployed with containers and automated experiment tracking, improving reproducibility and reducing setup time for new runs.

Co-Op: AI for Drug Discovery, Pfizer, Cambridge, MA *September 2024 – Dec 2024*

- Accelerated and optimized antibody sequence search and global alignment through protein large language models.
- Built protein vector store database through LlamaIndex and proposed germline filtering mechanism for acceleration.

Research Assistant under Prof. Howie Choset, Carnegie Mellon University *July 2024 – September 2024*

- Developed software for transforming 2D PCB design onto arbitrary high curvature 3D surface for 3D printing.
- Implemented proposed circuit mapping framework on aerosol jet additive manufacturing machine to preserve properties.

Research Assistant under Professor Ziming Zhang, WPI, MA *December 2023 – June 2024*

- Modeled RNN hidden-state evolution as a conditional probability path under linear diffusion priors, yielding closed-form supervision at every timestep; decoupled the sequential dependency of backpropagation through time to enable parallel training, stable gradients over long horizons, and inference from partial input sequences. Implemented text-to-image AI on large-scale datasets for synthetic identity generation for face recognition training.
- Explored the use of pre-defined Extreme Value Distribution in self-supervised learning to avoid dimension collapse.

Research Student under Prof. Feng Tian, Harvard University @ BIDMC *September 2023 – May 2024*

- Incorporated GPT to create embeddings for scRNA sequencing and text data for gene functionality prediction and unsupervised cell clustering.
- Visualized single cell sequencing bioinformatics data using Seurat package in RStudio.

Research Assistant under Professor Markus P. Nemitz, WPI, MA *April 2023 – December 2023*

- Designed a closed-loop FDM printing system optimized for real-time detection and remediation of permeability-related defects and formulated a software architecture capable of executing layer-wise defect detection and correction.
- Prototyped and developed a design of parameterized soft pneumatic assistive exoskeletons that is printable via standard FDM for personalized, reliable assistive devices that are affordable and easy to make.

R&D Co-Op, Olympus Corporation of the Americas, Westborough, MA *January 2023 – July 2023*

- Developed new and incorporated machine learning models to differentiate between human tissue, endoscope, and stones with laser spectral data, enabling smart features for the medical device; achieved nearly 100% sensitivity and specificity on all testing data under different data collection conditions.
- Implemented data logger processor program with tkinter and seaborn to filter and visualize data from thermocouples.

Research Assistant under Professor Jacob Richard Whitehill, WPI, MA *May 2022 – January 2023*

- Proposed and implemented an automatic and semi-automatic negativity filter and a classroom emotions analytics system using Speech Emotion Recognition, Google Cloud NLP APIs, and face emotions detection on OpenCV.
- Utilized Pandas and NumPy to achieve an AUC score of 83.3% with semi-automatic system on ∼1000GB datasets.

Awards/Honors

Amazon Fulfillment Technologies and Robotics Day One Fellowship *September 2025*
WPI Presidential Scholarship (full four years) *August 2021 – May 2025*
MIT Math Prize for Girls *October 2021*
American Mathematics Competition: National top 5% *January 2021*

Selected Projects

AI Agent for Alzheimer's *March 2026 – Present*

- Developing an AI-driven framework through CrewAI that aligns multi-modal sensor and health record data to enable early detection of Alzheimer's disease.

AI Bomberman Bot *January 2024 – March 2024*

- Devised an algorithm combining A* and reinforcement learning for path-planning of a simulated character to earn a higher score and escape from the world map. Interactive features and full graphics are provided.

Speaker Recognition System *July 2023 – Aug 2023*

- Implemented and modified SpeechBrain into a real-time speech activity and speaker recognition system based on PyTorch with user interface to process identification every second based on existing voice samples; system achieved 86.7% accuracy.
- Enabled a client-server data transfer pipeline to accelerate the speech processing speed from ∼1.5 seconds to ∼0.14 seconds.

Multi-thread Sport Scheduling System *September 2022 – October 2022*

- Programmed a multi-threaded sports park simulation system in C that schedules sports from multiple users concurrently.
- Implemented multiple mutexes and condition variables to protect shared resources and prevent race conditions.

Technical Skills

Programming Languages: Python (most proficient), C/C++, Java, R, Shell Scripting

ML Frameworks: PyTorch, HuggingFace Transformers, TensorFlow, OpenCV, LlamaIndex, CrewAI

ML Methods: Foundation model pretraining (Transformer, Mamba, LLMs), distributed training (HPC, AWS), retrieval systems & vector databases (RAG), reinforcement learning, diffusion models, VLM/VLA models, world models, self-supervised learning, multi-agent systems

Data & Infrastructure: SQL, Pandas, NumPy, Matplotlib, Seaborn, Docker, Git, Linux, AWS, Google Cloud Platform, ROS2, Conda, experiment tracking